JAVA OOP LABORATORY EXERCISE

CAMPUS CAFÉ ORDERING SYSTEM

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Exercise Title:	Campus Café Ordering System

WRITTEN REPORT

This project is a simple Java console-based ordering system for a campus café. It allows the program to display customer orders, calculate item prices, apply discounts or vouchers, and show the final amount to be paid. It also demonstrates how object-oriented programming can organize a program into reusable and understandable parts.

Concept Explanation

Classes and Objects: Classes serve as blueprints for items and orders, while objects represent actual menu items and customer orders created while the program runs.

Constructors: Constructors initialize objects by assigning their starting values, such as an item's name, code, and base price.

Encapsulation: Encapsulation protects an object's data by keeping fields private and providing methods to access or update them safely.

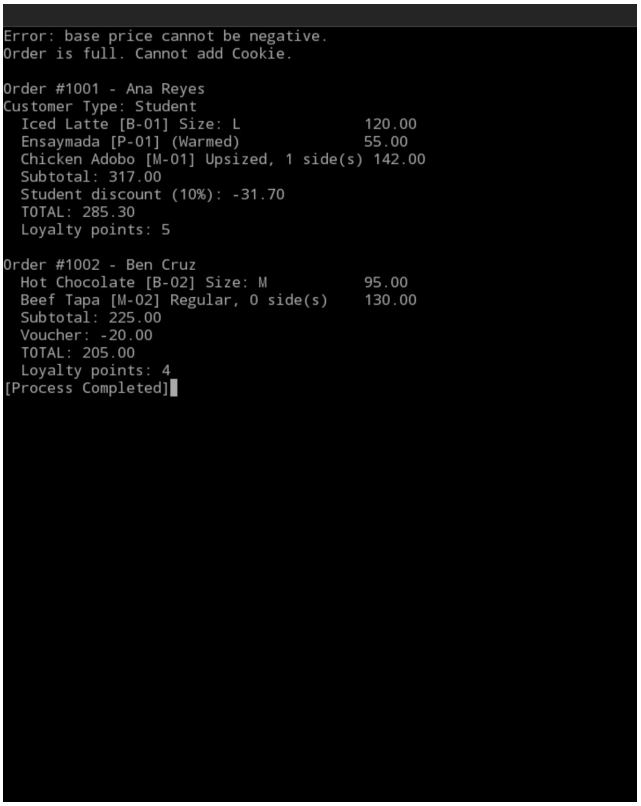
Inheritance: Inheritance allows a specialized class to reuse fields and methods from a parent class, reducing repeated code.

Polymorphism: Polymorphism allows different menu-item classes to provide their own version of a method, such as computing the final price.

Abstraction: Abstraction shows the important operations while hiding implementation details. It can be used to define a common structure for menu items.

Program Output

The screenshot below shows the sample program output. It includes validation messages, two customer orders, item prices, subtotal, discount or voucher, total amount, and loyalty points.



Challenges Encountered and Solutions

One challenge was handling invalid values and preventing the program from accepting an order when the order limit had already been reached. These were addressed through validation checks and clear error messages. Another challenge was making sure that prices, discounts, and totals were calculated correctly. The program computes the subtotal first, then applies the appropriate discount or voucher before displaying the final total.

Conclusion: This exercise helped demonstrate how Java OOP concepts work together in a practical café ordering program. It also provided practice in organizing code, validating input, and calculating customer payments.